Lec 6. Point Estimation

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Contents

6.1 Estimation principle 1
6.1.1 Method of Moments, MOM 6
6.1.2 Maximum Likelihood Estimators (MLEs) 10
6.1.3 Computing the MLE 11
6.2 Minimal contrast estimation and estimating equations 34

Learning objectives

After this section, you should be able to ...

- understand the criteria of point estimation
- understand the popular estimation methods
- find the estimator and its properties

6.1 Estimation principle

A point estimator uses the information available in a sample to obtain a single number that estimates a population parameter.

Definition

Let $X = (X_1, \ldots, X_n) \in \mathcal{X}$ be a sample from the population $F_\theta, \theta \in \Theta \subset R^d$, then **any statistic** $\hat{g}_n(X) = \hat{g}(X_1, \ldots, X_n)$ is said to be an estimator of parameter of interest $g(\theta)$.

Suppose we want to estimate θ, how do we find a function $\hat{\theta}_n(X)$ of the sample X that in some sense "is close" to the unknown θ?

Small sample properties

- The criterion of unbiasedness

Definition

An estimator $\hat{g}_n(X)$ is an unbiased estimator (无偏估计量) of $g(\theta)$, if

$$E_\theta \hat{g}_n(X) = g(\theta), \ \forall \theta \in \Theta$$

The bias (偏差)of estimator $\hat{g}_n(X)$ is defined as $E_\theta \hat{g}_n(X) - g(\theta)$. $\hat{g}_n(X)$ is said to be asymptotically unbiased if its bias goes to zero as the sample size $n \to \infty$.

Small sample properties

- Relative efficiency

Definition

Given two different unbiased estimators $\hat{g}_n(X)$ and $\tilde{g}_n(X)$ of $g(\theta)$, $\hat{g}_n(X)$ is said to be relatively more efficient than $\tilde{g}_n(X)$ if

$$Var_\theta(\tilde{g}_n(X)) \geq Var_\theta(\hat{g}_n(X), \ \forall \theta \in \Theta$$

and '>' holds for at least one value of θ. The efficiency of $\hat{g}_n(X)$ relative to $\tilde{g}_n(X)$, denoted $eff(\hat{g}_n, \tilde{g}_n)$, is given by

$$eff(\hat{g}_n, \tilde{g}_n) = \frac{Var_\theta(\tilde{g}_n(X))}{Var_\theta(\hat{g}_n(X))}$$

- **Mean square error (MSE):** for an estimator $\hat{g}_n(X)$ of $g(\theta)$:

$$MSE(\hat{g}_n(X)) = E(\hat{g}_n(X) - g(\theta))^2 = Var(\hat{g}_n(X)) + (E\hat{g}_n(X) - g(\theta))^2$$

Large sample properties

- The criterion of consistency

Definition

An estimator $\hat{g}_n(X)$ is a consistent estimator (相合估计量) of $g(\theta)$, if $\hat{g}_n(X) \xrightarrow{p} g(\theta)$, i.e., if $\hat{g}_n(X)$ converges in probability to $g(\theta)$. Otherwise $\hat{g}_n(X)$ is said to be inconsistent.

- Strongly consistent estimator: $\hat{g}_n(X) \xrightarrow{a.s.} g(\theta)$
- Consistent estimator in r-mean: $\lim_{n\to\infty} E|\hat{g}_n(X) - g(\theta)|^r = 0$.

Large sample properties

- Asymptotical normality

Definition

We say an estimator $\hat{g}_n(X)$ of $g(\theta)$ has asymptotical normality property, if

$$B_n^{-1}(\theta)[\hat{g}_n(X) - g(\theta)] \xrightarrow{\mathcal{L}} N(0, I)$$

for some positive series $B_n(\theta)$ as $n \to \infty$.

6.1.1 Method of Moments, MOM

The method of moments (矩估计方法) is a very simple (plug-in or substitution) procedure for finding an estimator for one or more parameters of a statistical model. It is one of the oldest methods for deriving point estimators.

Definition

If $g(\theta) = G(\alpha_1, \ldots, \alpha_k; \mu_2, \ldots, \mu_s)$, i.e., a function of popula- tion raw and central moments, then the method of moments estimator for $g(\theta)$ is defined as

$$\hat{g}_n(X) = G(a_{n1}, \ldots, a_{nk}; m_{n2}, \ldots, m_{ns})$$

where $a_{n1}, \ldots, a_{nk}; m_{n2}, \ldots, m_{ns}$ are sample moments of corresponding population moments.

Plug-in principle: $\hat{\theta}(X) = \theta(F_n)$ is a plug-in estimate of $\theta(F)$ as F_n is an estimate of F.

Examples

Properties

Theorem 1. *$\hat{g}_n(X) = G(a_{n1}, \ldots, a_{nk}; m_{n2}, \ldots, m_{ns})$ is a consistent estimator for $g(\theta) = G(\alpha_1, \ldots, \alpha_k; \mu_2, \ldots, \mu_s)$ if G is continuous function.*

Theorem 2. *$\hat{g}_n(X) = G(a_{n1}, \ldots, a_{nk}; m_{n2}, \ldots, m_{ns})$ is an asymptotically normal estimator for $g(\theta) = G(\alpha_1, \ldots, \alpha_k; \mu_2, \ldots, \mu_s)$ if G is continuously differentiable.*

6.1.2 Maximum Likelihood Estimators (MLEs)

Definition

Let $f(x, \theta)$ be the "probability function" of a sample $X = (X_1, \ldots, X_n)$ with parameter space Θ. If $X = x$ is observed, then

$$L_n(\theta) \equiv L_n(\theta|x) = f(x, \theta)$$

is called the likelihood function (似然函数). For each sample point $x = (x_1, \ldots, x_n)$, let

$$\hat{\theta}(x) = \arg\max_{\theta \in \Theta} L_n(\theta)$$

Then $\hat{\theta}(X)$ is called the maximum likelihood estimator (MLE, 最大似然估计量) of the parameter θ.

6.1.3 Computing the MLE

- Closed-form MLEs: $\hat{\theta} = \arg\max L(\theta)$
  - monotone likelihood
  - smooth likelihood: working on <u>log-likelihood</u>
    (1) find the roots of $\partial\ell(\theta)/\partial\theta = 0$, where $\ell(\theta) = logL(\theta)$;
    (2) find/verify the maxima
- Computational methods:
  - Gradient/Newton methods

$$\theta^{(t+1)} = \theta^{(t)} + \Delta \left.\frac{\partial \ell(\theta)}{\partial \theta}\right|_{\theta=\theta^{(t)}}, \text{ where } \Delta > 0 \text{ is a step size.}$$

  - Expectation-Maximization Algorithm (EM algorithm)

Examples

EM Algorithm

In many problems MLE based on observed data X would be greatly simplified if we had additionally observed another piece of data Y. Y is called the hidden or latent data.

Let $\ell(\theta) = \log f(x|\theta)$ be the observed log-likelihood and also define the complete data log-likelihood:

$$\begin{aligned}\ell_c(\theta) &= \log f(x, y|\theta) = \log f(y|x, \theta) f(x|\theta) \\ &= \log f(y|x, \theta) + \log f(x|\theta) = \log f(y|x, \theta) + \ell(\theta)\end{aligned}$$

Suppose our current guess of θ is $\theta^{(t)}$ and that we would like to imporve this guess. Consider

$$\ell(\theta) - \ell(\theta^{(t)}) = \ell_c(\theta) - \ell_c(\theta^{(t)}) + \log \frac{f(y|x, \theta^{(t)})}{f(y|x, \theta)}$$

Now take expectation of both sides with respect to $y \sim f(y|x, \theta^{(t)})$, we have:

$$\ell(\theta) - \ell(\theta^{(t)}) = \mathbb{E}_y[\ell_c(\theta)] - \mathbb{E}_y[\ell_c(\theta^{(t)})] + D(f(y|x, \theta^{(t)}) \| f(y|x, \theta))$$

Since $D(f(y|x,\theta^{(t)})\|f(y|x,\theta)) \geq 0$, we have the following inequality:

$$\ell(\theta) - \ell(\theta^{(t)}) \geq \mathbb{E}_y[\ell_c(\theta)] - \mathbb{E}_y[\ell_c(\theta^{(t)})] = Q(\theta,\theta^{(t)}) - Q(\theta^{(t)},\theta^{(t)})$$

where

$$Q(\theta,\theta') := \mathbb{E}_{f(y|x,\theta')}[\log f(x,y|\theta)]$$

is the expectation of complete data log-likelihood.

We choose $\theta^{(t+1)}$ as the solution of the following optimization problem:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta,\theta^{(t)})$$

The process of EM algorithm is as follows:

for $\mathrm{t} = 0, 1, 2, \ldots$

- **Init:** $\mathrm{t} = 0, \theta^{(0)} = 0$ or random value
- **E step:**

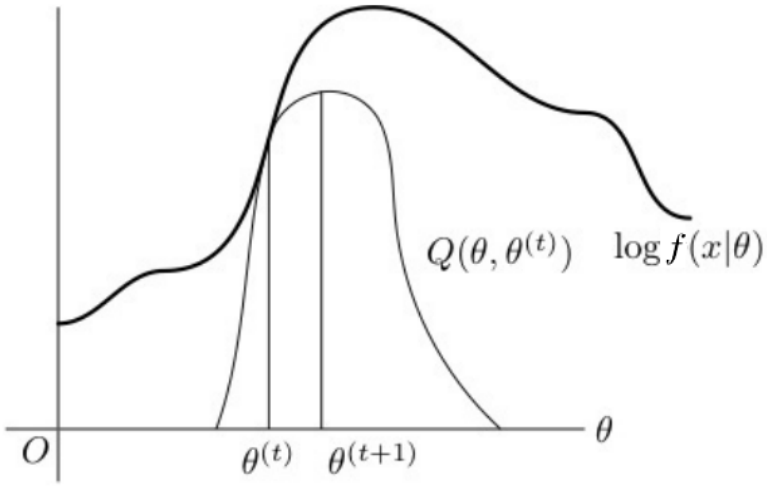
$$Q(\theta,\theta^{(t)}) = \mathbb{E}_{f(y|x,\theta^{(t)})}[\log f(x,y|\theta)]$$

- **M step:**

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta,\theta^{(t)})$$

The E-step and M-step repeat until convergence.

The EM algorithm is an attractive option if the Q function is easily computed and optimized. The relationship between $\log f(x, \theta), Q(\theta, \theta^{(t)}), \theta^t$ and $\theta^{(t+1)}$ are depicted in the following figure:



The two key properties of the EM algorithm are:

1. $\log f(x|\theta^{(0)}) \leq \log f(x|\theta^{(1)}) \leq \ldots$
2. It converges to stationary point(e.g. local max)

Example

(Mixture of normal) Suppose:

$$X_1, X_2, \ldots, X_n \overset{iid}{\sim} \sum_{j=1}^{m} \pi_j N\left(\mu_j, \sigma_j^2\right)$$

find the MLE $(\hat{\mu}_j, \hat{\sigma}_j^2), j = 1, \ldots, m$.

Let $y_1, \ldots, y_n$ be the latent member index, then

$$f(x, y \mid \theta) = \prod_{i=1}^{n} \sum_{j=1}^{m} \pi_j \frac{1}{\sqrt{2\pi}\sigma_j} \exp\{-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}\} 1_{y_i = j}$$

Thus,

$$\log f(x, y \mid \theta) = \sum_{i=1}^{n} \sum_{j=1}^{m} \log\left(\frac{\pi_j}{\sqrt{2\pi}\sigma_j} \exp\{-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}\}\right) 1_{y_i = j}$$

and

$$
\begin{aligned}
&\mathbb{E}_{f(y|x,\theta^{(t)}}[\log f(x,y \mid \theta)] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} \log(\frac{\pi_j}{\sqrt{2\pi}\sigma_j}\exp\{-\frac{(x_i-\mu_j)^2}{2\sigma_j^2}\})\mathbb{E}_{f(y|x,\theta^{(t)})}[1_{y_i=j}] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} \log\left(\frac{\pi_j}{\sqrt{2\pi}\sigma_j} exp\{-\frac{(x_i-\mu_j)^2}{2\sigma_j^2}\}\right)\frac{\pi_j^{(t)}N(x_i;\mu_j^{(t)},(\sigma_j^{(t)})^2)}{\sum_{l=1}^{m} p_l^{(t)}N(x_i;\mu_l^{(t)},(\sigma_l^{(t)})^2)}.
\end{aligned}
$$

Denote $f^{(t)}(y_i = j) = \frac{\pi_j^{(t)}N(x_i;\mu_j^{(t)},(\sigma_j^{(t)})^2)}{\sum_{l=1}^{m} p_l^{(t)}N(x_i;\mu_l^{(t)},(\sigma_l^{(t)})^2)}$, we have the expression of $Q(\theta,\theta^{(t)})$:

$$
\begin{aligned}
Q(\theta,\theta^{(t)}) &= \sum_{i=1}^{n}\sum_{j=1}^{m} f^{(t)}(y_i=j)\log(\pi_j^{(t)}N(x_i;\mu_j,\sigma_j^2)) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} f^{(t)}(y_i=j)\log(N(x_i;\mu_j,\sigma_j^2)) + \text{ constant}
\end{aligned}
$$

Set $\frac{\partial Q}{\partial \theta} = 0$, we have:

$$\mu_j^{(t+1)} = \frac{\sum_{i=1}^n f^{(t)}(y_i = j) x_i}{\sum_{i=1}^n f^{(t)}(y_i = j)}$$

$$(\sigma_j^{(t+1)})^2 = \frac{\sum_{i=1}^n (x_i - \mu_j^{(t+1)})^2 f^{(t)}(y_i = j)}{\sum_{i=1}^n f^{(t)}(y_i = j)}$$

$$\pi_j^{(t+1)} = \frac{\sum_{i=1}^n f^{(t)}(y_i = j)}{\sum_{j=1}^m \sum_{i=1}^n f^{(t)}(y_i = j)}$$

Properties of MLEs

we want to show that the MLE $\hat{\theta}(X)$ is often

- functional invariant, $h(\hat{\theta})$ is the MLE of $h(\theta)$
- function of every sufficient statistic
- consistent, $\hat{\theta}(X) \xrightarrow{P} \theta_0$
- asymptotically normal, $\sqrt{n}\left(\hat{\theta}(X) - \theta_0\right) \overset{\mathcal{D}(\theta_0)}{\rightarrow}$ Normal R.V.
- asymptotically efficient, i.e., if we want to estimate θ_0 by any other estimator within a "reasonable class," the MLE is the most precise (will discuss in Lec 7).

Theorem 3 (**Invariance Principle of MLEs**). *If $\hat{\theta}$ is the MLE of θ, then $h(\hat{\theta})$ is the MLE of $h(\theta)$ where h is a function with domain Θ.*

Proof. When h is one to one, let $\eta = h(\theta)$. Then the inverse function h^{-1} exists and $\theta = h^{-1}(\eta)$. Hence

$$f(x|\theta) = f(x|h^{-1}(\eta))$$

is the joint pdf or pmf of x. So the likelihood function of $h(\theta) = \eta$ is

$$L^*(\eta) \equiv L(h^{-1}(\eta)).$$

Also note that

$$\sup_{\eta} L^*(\eta) = \sup_{\eta} L(h^{-1}(\eta)) = L(\hat{\theta})$$

Thus

$$\hat{\eta} = h(\hat{\theta})$$

is the MLE of $\eta = h(\theta)$ when h is one to one.

If $h(\theta)$ is not one-to-one, then for a given value η there may be more than one value of θ that satisfies $h(\theta) = \eta$. In such cases, the correspondence between the maximization over η and that over θ can break down. For example, if $\hat{\theta}$ is the MLE of θ, there may be another value of θ, say θ_0, for which $h(\hat{\theta}) = h(\theta_0)$. We need to avoid such difficulties. We proceed by defining for $h(\theta)$ the induced likelihood function L^*, given by

$$L^*(\eta|\mathbf{x}) = \sup_{\{\theta : h(\theta) = \eta\}} L(\theta|\mathbf{x})$$

The value $\hat{\eta}$ that maximizes $L^*(\eta|\mathbf{x})$ will be called the MLE of $\eta = h(\theta)$.

Let $\hat{\eta}$ denote the value that maximizes $L^*(\eta|\mathbf{x})$. We must show that $L^*(\hat{\eta}|\mathbf{x}) = L^*[h(\hat{\theta})|\mathbf{x}]$. Now, as stated above, the maxima of L and L^* coincide, so we have

$$\begin{aligned} L^*(\hat{\eta}|\mathbf{x}) &= \sup_{\eta} \sup_{\{\theta : h(\theta) = \eta\}} L(\theta|\mathbf{x}) && \text{(definition of } L^*\text{)} \\ &= \sup_{\theta} L(\theta|\mathbf{x}) \\ &= L(\hat{\theta}|\mathbf{x}), && \text{(definition of } \hat{\theta}\text{)} \end{aligned}$$

where the second equality follows because the iterated maximization is equal

to the unconditional maximization over θ, which is attained at $\hat{\theta}$. Furthermore

$$\begin{aligned} L(\hat{\theta}|\mathbf{x}) &= \sup_{\{\theta: h(\theta)=h(\hat{\theta})\}} L(\theta|\mathbf{x}) && (\hat{\theta} \text{ is the MLE }) \\ &= L^*[h(\hat{\theta})|\mathbf{x}]. && (\text{definition of } L^*) \end{aligned}$$

Hence, the string of equalities shows that $L^*(\hat{\eta}|\mathbf{x}) = L^*(h(\hat{\theta})|\mathbf{x})$ and that $h(\hat{\theta})$ is the MLE of $h(\theta)$. $\square$

Note: The invariance property of MLEs also holds in the multivariate case.

Theorem 4 (**Existence and uniqueness**). *Let* $\mathbf{X}_n = (X_1, \ldots, X_n)$ *be a simple sample from a population with density* $f(x, \theta), \theta \in \Theta$ *open* $\subset R^d$. *If further* $\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} lnf(x_i, \theta)$ *is strictly concave and* $\ell_n(\theta) \to -\infty$ *as* $\theta \to \partial\Theta$ *(the boundary of* Θ*), then the MLE* $\hat{\theta}(\mathbf{x}_n)$ *exists and is unique.*

Example

$X_1, \ldots, X_n i.i.d \sim B(1, p)$:

$$\ell_n(p) = log(p) \sum x_i + log(1-p)(n - \sum x_i)$$

Example

$X_1, \ldots, X_n i.i.d \sim U(0, \theta)$:

$$L_n(\theta) = \frac{1}{\theta^n} I(0 < x_{(n)} < \theta)$$

The MLE of θ defined by $\arg\max$ does **not exist**, but a remedy is to define the MLE by

$$\hat{\theta}(x) = \arg\sup_{\theta \in \Theta} L(\theta)$$

Theorem 5. *Suppose random sample $X = (X_1, \ldots, X_n)$ is sampled from a canonical full rank exponential family, i.e., the density/pmf function is given by*

$$f(x, \theta) = h(x) \exp[\theta^T T(x) - A(\theta)], \text{ for } x \in \mathcal{X}$$

if

$$\frac{\partial A(\theta)}{\partial \theta} = \frac{1}{n} \sum_{i=1}^{n} T(X_i)$$

has a solution in Θ^o, then it must be the unique MLE of θ.

Theorem 6 (**MLE consistency**). *Let $X = (X_1, \ldots, X_n)$ be a random sample from a population with density $f(x, \theta), \theta \in \Theta$ open $\subset R^d$. Assume that (1) θ is identifiable, i.e. for any $\theta' \neq \theta$, there exists x such that $f(x, \theta') \neq f(x, \theta)$; (2) the support of $f(\cdot, \theta)$ does not depend on θ, and $f(x, \theta)$ is differentiable with respect to θ in Θ; and (3) the true value θ_0 is an interior point of parameter space Θ. Then the likelihood has a maximizer $\hat{\theta}$ and $\hat{\theta} \to_p \theta_0$.*

Consistency of MLE

Denote by $\ell_n(\theta|x) = \sum_{i=1}^{n} log f(x_i, \theta)$.

Proof. We consider $d = 1$. Let $\delta > 0$ such that $(\theta_0 - \delta, \theta_0 + \delta) \subseteq \Theta \subset R$. Define,

$$A_{n,\delta} = \{X : \ell_n(\theta_0|X) > \ell_n(\theta_0 - \delta|X)\}.$$
$$B_{n,\delta} = \{X : \ell_n(\theta_0|X) > \ell_n(\theta_0 + \delta|X)\}.$$

where $X = (X_1, \ldots, X_n)$. Since $E_{\theta_0} log \frac{f(X,\theta_0 \pm \delta)}{f(X,\theta_0)} < 0$, by the strong LLN we know $P(A_n), P(B_n)$ approach 1. Let $S_{n,\delta} = A_{n,\delta} \cap B_{n,\delta}$. Hereby,

$$\begin{aligned}
1 &\geq P(S_{n,\delta}) \\
&= P(A_{n,\delta} \cap B_{n,\delta}) \\
&= 1 - P\left(A^c_{n,\delta} \cup B^c_{n,\delta}\right) \\
&\geq 1 - P\left(A^c_{n,\delta}\right) - P\left(B^c_{n,\delta}\right) \\
&= P(A_{n,\delta}) + P(B_{n,\delta}) - 1.
\end{aligned}$$

Subsequently,

$$1 \geq \lim_{n\to\infty} P(S_{n,\delta}) \geq \lim_{n\to\infty} P(A_{n,\delta}) + \lim_{n\to\infty} P(B_{n,\delta}) - 1.$$

Thus, $1 \geq \lim_{n\to\infty} P(S_{n,\delta}) \geq 1$ and, therefore, $\lim_{n\to\infty} P(S_{n,\delta}) = 1$.

Furthermore, let is prove that $S_{n,\delta} \subseteq \left\{X : \left|\hat{\theta}_n - \theta_0\right| < \delta\right\}$.

Assume that $X \in S_{n,\delta}$, then $\ell_n(\theta_0|X) > \ell_n(\theta_0 - \delta|X) \wedge \ell_n(\theta_0|X) > \ell_n(\theta_0 + \delta|X)$. Since $\ell_n(\theta|X)$ is differentiable on Θ, then, it is differentiable and, in consequence, continuous on $(\theta_0 - \delta, \theta_0 + \delta)$. Hence, $\ell_n(\theta|X)$ has a local maximum on this interval, $\hat{\theta}_n$, by hypothesis. In this manner,

$$\theta_0 - \delta < \hat{\theta}_n < \theta_0 + \delta \quad \Leftrightarrow \left|\hat{\theta}_n - \theta_0\right| < \delta.$$

This validates that $S_{n,\delta} \subseteq \left\{X : \left|\hat{\theta}_n - \theta_0\right| < \delta\right\}$. Thence,

$$1 = \lim_{n\to\infty} P(S_{n,\delta}) \leq \lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta_0\right| < \delta\right) \leq 1$$

As a result, $\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta_0\right| < \delta\right) = 1$, i.e., $\hat{\theta}_n \xrightarrow{p} \theta$. $\square$

Theorem 7 (**MLE asymptotic normality**). *In the setting above, assume that conditions (1)-(3) in the MLE consistency theorem hold. In addition, assume that (4) $f(x,\theta)$ is three differentiable with respect to θ and we can interchange integration with respect to x; and (5) $\left|\partial^3 \log f(x,\theta)/\partial\theta^3\right| \leq M(x)$ and $E[M(X_i)] < \infty$. Then*

$$\sqrt{n}\left(\hat{\theta}_n - \theta_0\right) \to N\left(0, \frac{1}{I(\theta_0)}\right)$$

in distribution.

Denote by $\ell_n(\theta) = \sum_{i=1}^{n} logf(x_i, \theta)$.

Proof. The first step is to expand $\ell_n'(\hat{\theta}_n)$ in a power series around θ_0 :

$$\ell_n'(\hat{\theta}_n) = \ell_n'(\theta_0) + (\hat{\theta}_n - \theta_0)\ell_n''(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)^2 \ell_n'''(\theta_n^*)$$

for some θ_n^* between $\hat{\theta}_n$ and θ_0. Thus,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{\sqrt{n}\{\ell_n'(\hat{\theta}_n) - \ell_n'(\theta_0)\}}{\ell_n''(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)\ell_n'''(\theta_n^*)}$$

Because $\ell_n'(\hat{\theta}_n) = 0$. Noting that

$$\frac{1}{\sqrt{n}}\ell_n'(\theta_0) = \sqrt{n}(\frac{1}{n}\sum_{i=1}^{n}\frac{d}{d\theta}\log f_{\theta_0}(X_i)) \xrightarrow{\mathcal{L}} N\{0, I(\theta_0)\}$$

by the central limit theorem, and

$$\frac{1}{n}\ell_n''(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}\frac{d^2}{d\theta^2}\log f_{\theta_0}(X_i) \xrightarrow{P} -I(\theta_0)$$

by the weak law of large numbers. Finally, we would like to have the term involving $\ell_n'''(\theta_n^*)$ disappear, so clearly it is enough to show that $\frac{1}{n}\ell_n'''(\theta)$ is bounded in probability in a neighborhood of θ_0. $\square$

Different likelihoods

Let the parameter be (β, θ) and θ is not of interest, then the standard likelihood by $L(\beta, \theta|x)$.

- **Maximum Likelihood:** Find β and θ that maximizes $L(\beta, \theta|x)$.
- **Profile Likelihood:** If we can express θ as a function of β then we replace θ with the corresponding function. Say, $\theta = g(\beta)$. Then, we maximize:

  $$L(\beta, g(\beta)|x)$$

  Usually, the profile likelihood estimate $\hat{\beta}$ can be done in the following two steps:

  - fixed β, find

    $$g(\beta) = \arg\max_{\theta} L(\beta, \theta|x)$$

  - find $\hat{\beta}$ such that

    $$\hat{\beta} = \arg\max L(\beta, g(\beta)|x)$$

- **Partial likelihood** If we can write the likelihood function as:

$$L(\beta, \theta|x) = L_1(\beta|x)L_2(\theta|x)$$

  Then we simply maximize $L_1(\beta|x)$.

- **Induced likelihood** If $\eta = h(\theta)$, then

$$L^*(\eta) \equiv \sup_{\theta:h(\theta)=\eta} L(\theta)$$

6.2 Minimal contrast estimation and estimating equations

We consider a function that we shall call a contrast (discrepancy) function:

$$\rho : \mathcal{X} \times \Theta \to R$$

and define $\mathcal{D}(\theta_0, \theta)$ to measure the discrepancy between θ and θ_0(true value):

$$D(\theta_0, \theta) \equiv E_{\theta_0} \rho(X, \theta)$$

- We **require** that $D(\theta_0, \theta)$ is uniquely minimized for $\theta = \theta_0$.
- Usually, $\rho(X, \theta) = \frac{1}{n} \sum_{i=1}^{n} \rho(X_i, \theta) \longrightarrow D(\theta_0, \theta)$ (as $n \longrightarrow \infty$). It is natural to consider $\widehat{\theta}(X)$ minimizing $\rho(X, \theta)$.
- Now suppose the true θ_0 is an interior point of Θ, and $\theta \to D(\theta_0, \theta)$ is smooth. Then we expect

$$\nabla_\theta D(\theta_0, \theta) = 0$$

where ∇ denotes the gradient $\nabla_\theta = \left(\frac{\partial}{\partial \theta_1}, \ldots, \frac{\partial}{\partial \theta_d} \right)'$

- Arguing heuristically again we are led to estimates $\widehat{\theta}$ that solve

$$\nabla_\theta \rho(\mathbf{X}, \widehat{\theta}) = 0$$

which is a special form of following estimating equations.

- **Estimation equation:** More generally, suppose we are given a function $\Psi : \mathcal{X} \times R^d \to R^d, \Psi \equiv (\psi_1, \ldots, \psi_d)^T$ and define

$$S(\theta_0, \theta) = E_{\theta_0} \Psi(X, \theta)$$

Suppose $S(\theta_0, \theta) = 0$ has $\theta = \theta_0$ as its unique solution for all $\theta_0 \in \Theta$. Then we say $\widehat{\theta}$ solving

$$\Psi(X, \hat{\theta}) = 0$$

is an estimating equation estimate.

Least-square estimator

Example

Let $(\mathbf{X}_i, Y_i)$ be i.i.d. from

$$\begin{aligned} Y_i &= g\left(\mathbf{X}_i, \beta\right) + \varepsilon_i \\ &= \mathbf{X}_i^T \beta + \varepsilon_i, \qquad \text{if linear model} \end{aligned}$$

By letting

$$\rho(\mathbf{X}, \beta) = \sum_{i=1}^{n} \left[Y_i - g\left(\mathbf{X}_i, \beta\right)\right]^2$$

be a contract function, we have

$$\begin{aligned} D\left(\beta_0, \beta\right) &= E_{\beta_0} \rho(\mathbf{X}, \beta) \\ &= n E_{\beta_0} \left[Y - g(\mathbf{X}, \beta)\right]^2 \\ &= n E \left\{g\left(\mathbf{X}, \beta_0\right) - g(\mathbf{X}, \beta)\right\}^2 + n\sigma^2 \end{aligned}$$

which is indeed minimized at $\beta = \beta_0$. Hence, the minimum contrast estimator

is

$$\widehat{\beta} = \arg\min_{\beta} \sum_{i=1}^{n} \left[Y_i - g\left(\mathbf{X}_i, \beta\right)\right]^2 \text{ - least-squares.}$$

It satisfies the system of equations

$$\sum_{i=1}^{n} \left(Y_i - g\left(\mathbf{X}_i, \widehat{\beta}\right)\right) \frac{\partial g\left(\mathbf{X}_i, \beta\right)}{\partial \beta_j} = 0, j = 1, \cdots, d,$$

under some mild regularity conditions. One can easily check that $\psi_j(\beta) = (Y_i - g\left(\mathbf{X}_i, \beta\right)) \frac{\partial g(\mathbf{X}_i, \beta)}{\partial \beta_j}$ atisfies

$$E_{\beta_0} \psi_j(\beta)\Big|_{\beta=\beta_0} = E\left\{g\left(\mathbf{X}_i, \beta_0\right) - g\left(\mathbf{X}_i, \beta_0\right)\right\} \frac{\partial g\left(\mathbf{X}_i, \beta_0\right)}{\partial \beta_j} = 0$$

Thus, it is also an estimator based on the estimating equations.

L_1-regression

Consider

$$\rho(\mathbf{X}, Y, \beta) = \left| Y - \mathbf{X}^T \beta \right|.$$

Then,

$$D(\beta_0, \beta) = E_{\beta_0} \left| Y - \mathbf{X}^T \beta \right| = E \left| \mathbf{X}^T (\beta - \beta_0) + \varepsilon \right|.$$

For any a, define

$$f(a) = E|\varepsilon + a|.$$

Then,

$$\begin{aligned} f'(a) &= E \operatorname{sgn}(\varepsilon + a) \\ &= P(\varepsilon + a > 0) - P(\varepsilon + a < 0) \\ &= 2P(\varepsilon + a > 0) - 1 \end{aligned}$$

If $\operatorname{med}(\varepsilon) = 0$, then $f'(0) = 0$. In other words, $f(a)$ is minimized at $a = 0$, or $D(\beta_0, \beta)$ is minimized at $\beta = \beta_0$! Thus, if $\operatorname{med}(\varepsilon) = 0$, then

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \left| Y_i - \mathbf{X}_i^T \beta \right|.$$

is a minimum contrast estimator.

MLE as a minimum contrast estimator

Example

Let $L(\theta) = f(x, \theta)$ be the likelihood function, θ_0 is the true parameter.

Then, let $D(\theta_0, \theta) = -E_{\theta_0} \log f(X|\theta)$ and to say that $D(\theta_0, \theta)$ is minimized uniquely when $P_\theta = P_{\theta_0}$ is equivalent to

$$\begin{aligned} D(\theta_0, \theta) - D(\theta_0, \theta_0) &= -\left(E_{\theta_0} \log f(X|\theta) - E_{\theta_0} \log f(X|\theta_0)\right) \\ &= -E_{\theta_0} \log \frac{f(X|\theta)}{f(X|\theta_0)} > 0 \end{aligned}$$

unless $\theta = \theta_0$. Here $D(\theta_0, \theta_0)$ is called the entropy of X. In the case of $X_1, \ldots, X_n$ i.i.d.

$$\rho(X, \theta) = -\frac{1}{n} \sum \log f(X_i, \theta)$$

satisfies the condition of being a contrast function, and the MLE is a minimum contrast estimate.

Define the mutual entropy or Kullback-Leibler information divergence between $f(X|\theta_0)$ and $f(X|\theta)$ by

$$K\left(f\left(X|\theta_0\right), f\left(X|\theta\right)\right) = -E_{\theta_0} \log \frac{f(X|\theta)}{f\left(X|\theta_0\right)}$$

Then maximizing likelihood is equivalent to minimizing Kullback-Leibler divergence, i.e., maximizing the likelihood of data is equal to minimizing the difference between the estimate and the real data distribution.